



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
UNIT TEST
18-09-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]

Time : 3 Hours

PAPER-1

Maximum Marks : 183

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS
SECTION-I (i) : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

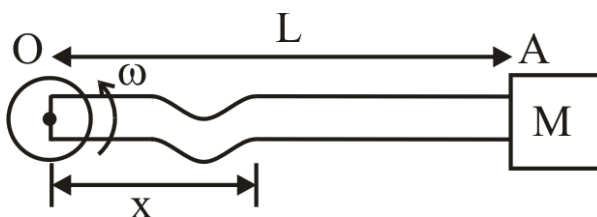
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

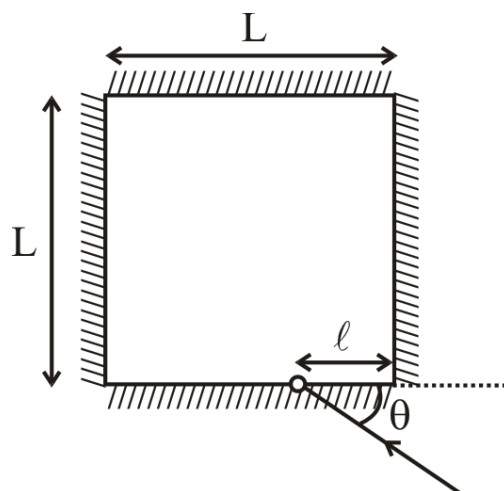
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. A block of mass M is connected to a uniform rope of length L having uniform mass per unit length μ . The other end of rope is hinged at O and system is rotating with constant angular velocity ω in gravity free space. A transverse pulse of wavelength λ is produced at O . Assume no loss in energy. Then (Given : $\mu = \frac{2M}{3L}$)

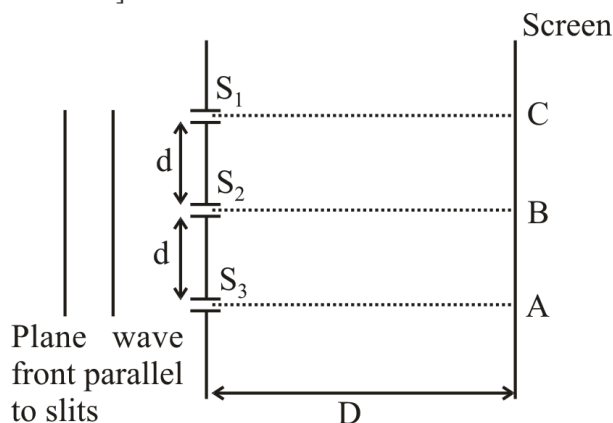


- (A) Wavelength of pulse decrease as it reaches point A and becomes $\frac{\sqrt{3}}{2}\lambda$.
- (B) Time taken by pulse to travel from O to A is $\frac{\pi}{3\sqrt{2}\omega}$.
- (C) As pulse propagate from O to A , its Amplitude continuously increases.
- (D) Pulse speed at point A in frame of ground is $\sqrt{\frac{5}{2}}\omega L$.

2. Four plane mirrors form a square with each side of length L . There is a small hole at one of the corners as shown in the figure. A ray of light passed through the hole at an angle θ and can come out of same hole. The cross-section of mirror configuration and the ray of light lies on the same plane. Which of the following statements are correct.



- (A) A ray of light will come out for $\theta = 45^\circ$ only if $l = \frac{L}{2}$.
- (B) A ray of light will come out for $\theta = 90^\circ$ for $0 < l < L$.
- (C) A ray of light will come out for $\theta = \tan^{-1} \left(\frac{1}{2n} \right)$, where n is a natural number for $l = \frac{L}{2}$.
- (D) A ray of light will come out for $\theta = 45^\circ$ for $0 < l < L$.
3. In given setup coherent light of wave length λ is used assuming $D \gg d$, select correct alternative [each slit produce same intensity on screen]



- (A) For $d = \sqrt{2\lambda D}$, intensity at C and A will be maximum but minimum at B.
- (B) For $d = 2\sqrt{\lambda D}$, intensity at A, B and C will be maximum intensity.
- (C) For $d = \sqrt{\frac{\lambda D}{2}}$, intensity at A, B and C will be $\left(\frac{5}{9}\right)^{\text{th}}$ maximum intensity.
- (D) For $d = \frac{\sqrt{\lambda D}}{2}$, intensity at A and C will be $\left(\frac{1}{9}\right)^{\text{th}}$ maximum intensity.

4. Two long parallel straight wire carry equal current I along positive z -direction. One of them intersect x -axis at point $(a, 0, 0)$ and other at $(-a, 0, 0)$. A third finite length wire of uniform mass per unit length λ carries current in negative z -direction and intersect y -axis at point $(0, b, 0)$. If acceleration due to gravity is $\vec{g} = -g\hat{j}$ and wire is suspended under gravitational and magnetic forces only then select correct alternatives.

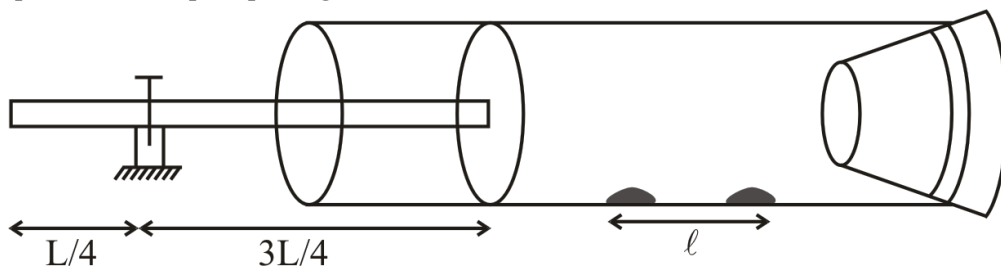
(A) Current in finite length wire is $\frac{\pi\lambda g(a^2 + b^2)}{\mu_0 Ib}$.

(B) When wire is displaced by Δy along y -axis and released then wire will perform SHM for $\Delta y \ll b$.

(C) Wire is displaced by Δy along y -direction and released then wire will perform SHM for $\Delta y \ll b$ and $b > a$.

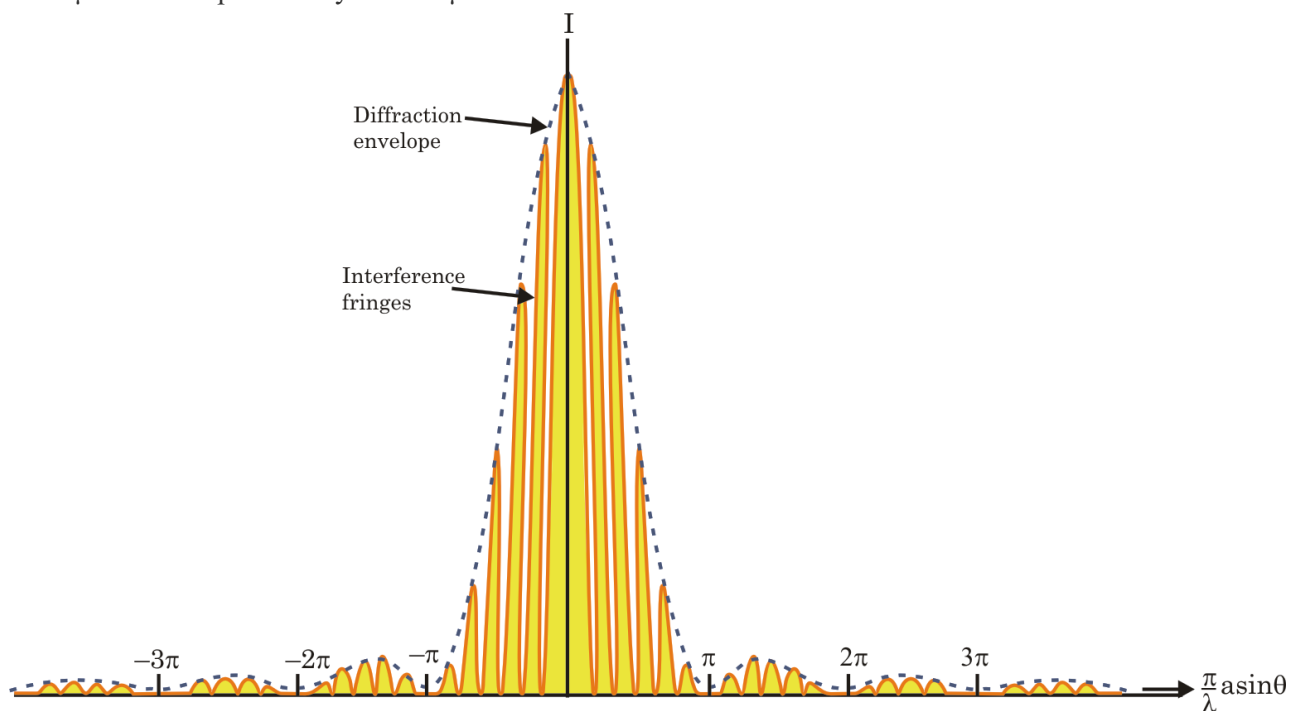
(D) Wire will perform SHM if displaced slightly in any direction perpendicular to z -axis.

5. A rod of length $L = 1\text{ m}$ is made of material having density $\rho = \frac{3g}{cc}$ and young's modulus $Y = 2.43 \times 10^9 \text{ N/m}^2$, is clamped at distance $\frac{L}{4}$ from one end and other end has as metal disk attached. Glass tube contain gas of density $\rho_0 = 0.01 \text{ g/cc}$ and bulk modulus $B_0 = 9 \times 10^5 \text{ N/m}^2$. When longitudinal oscillation are setup in rod then fine powder present in tube pill up at regular interval of $\ell = 3.33 \text{ cm}$, then select correct alternatives.



- (A) Speed of wave in rod is 900 m/s .
- (B) Speed of sound in gas 300 m/s
- (C) frequency of wave is nearly 4500 Hz .
- (D) Rod is vibrating in 2^{nd} overtone.
6. Consider an ideal solenoid having radius of cross section R . There are n turns/length. It carries a current of $I_0 \sin(\omega t)$. Consider a cylinder which is coaxial with the solenoid and has a length ℓ and radius r we want to find the rate of energy flow through this cylinder. ($r < R$) neglect magnetic field of induced electric field
- (A) The rate of energy flow through its flat face is zero.
- (B) At $t = \frac{\pi}{4\omega}$, energy is flowing outwards from the curved surface of the cylinder.
- (C) The rate of energy flow is proportional to $\mu_0 n^2 I_0^2 r^2 \omega \ell$.
- (D) At $t = \frac{\pi}{4\omega}$, rate of energy flow is maximum.

7. Consider a diffraction pattern of 2 slits as shown in figure. The pattern is formed by 2 slits of width $a = 3\mu\text{m}$ each separated by $d = 18\mu\text{m}$.



- (A) Here 6th order maxima of interference can't be seen.
- (B) If the wavelength is decreased, the central envelope becomes narrower.
- (C) If the wavelength is decreased, the width of each interference maxima becomes lesser.
- (D) If the wavelength is decreased, same number of interference maxima will be seen in the central envelope of diffraction.

SECTION-I (ii) : (Maximum Marks: 18)

- This section contains **SIX (6)** questions of matching type.
- This section contains **TWO** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : -1 In all other cases

Paragraph for Questions 8 - 10

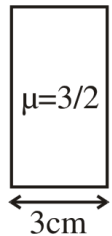
A charged particle is introduced at the origin ($x = 0, y = 0, z = 0$) with a given initial velocity $\vec{V}$. A uniform electric field $\vec{E}$ exists everywhere. The velocity $\vec{V}$, electric field $\vec{E}$ and magnetic field $\vec{B}$ are given in columns 1, 2 and 3 respectively. The quantities E_0, B_0 are positive in magnitude.

	Column-1		Column-2		Column-3
(I)	Positive charge with $\vec{V} = \frac{2E_0}{B_0} \hat{i}$.	(i)	$\vec{E} = E_0 \hat{j}$	(P)	$\vec{B} = -B_0 \hat{k}$
(II)	Negative charge with $\vec{V} = \frac{E_0}{B_0} (\hat{j} + \hat{k})$.	(ii)	$\vec{E} = E_0 \hat{i}$	(Q)	$\vec{B} = B_0 \hat{i}$
(III)	Positive charge with $\vec{V} = \frac{E_0}{B_0} (\hat{i} + \hat{j})$.	(iii)	$\vec{E} = E_0 \hat{k}$	(R)	$\vec{B} = -B_0 \hat{j}$
(IV)	Negative charge with $\vec{V} = \frac{E_0}{B_0} \hat{i}$.	(iv)	$\vec{E} = -E_0 \hat{i}$	(S)	$\vec{B} = B_0 \hat{k}$

- In which case particle will move on a helical path with decreasing pitch.
 (A) (IV), (ii), (Q) (B) (I), (iii), (R)
 (C) (II), (i), (R) (D) (III), (ii), (Q)
- In which case particle will move with velocity changing periodically.
 (A) (I), (iii), (R) (B) (IV), (i), (S)
 (C) (II), (iii), (S) (D) (III), (ii), (Q)
- In which case particle will move on a straight line with constant speed.
 (A) (I), (ii), (Q) (B) (I), (iii), (R)
 (C) (III), (ii), (S) (D) (IV), (i), (S)

Paragraph for Questions 11 - 13

Column-1 represent distance of real object from optical instrument-1 given in Column-2. 2nd optical instrument place in contact with 1st is given column-3.

	Column-1		Column-2		Column-3
(I)	15 cm	(i)	 $f = +10\text{cm}$	(P)	 Plane mirror
(II)	10 cm	(ii)	 $f = -10\text{cm}$	(Q)	 $R = 20\text{cm}$
(III)	20 cm	(iii)	 $f = 5\text{cm}$	(R)	 $R = 30\text{cm}$
(IV)	30 cm	(iv)	 $\mu = 3/2$ Plane slab 3cm	(S)	 $f = 30\text{cm}$

11. In which of following case final image will form on object itself.

(A) (II), (i), (P)	(B) (I), (i), (P)
(C) (I), (i), (R)	(D) (III), (iii), (R)
12. In which case distance between object and it's final virtual image will be 24 cm.

(A) (II), (iv), (P)	(B) (III), (ii), (Q)
(C) (IV), (iii), (S)	(D) (I), (i), (R)
13. In which case real image of same size as object forms.

(A) (I), (ii), (R)	(B) (II), (iii), (P)
(C) (III), (iv), (R)	(D) (IV), (ii), (S)

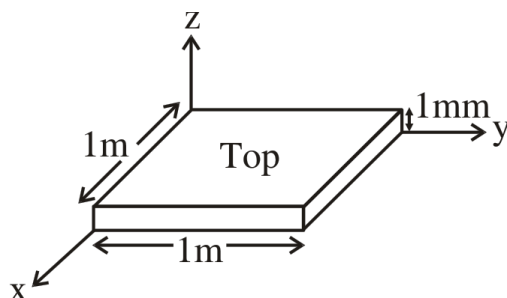
SECTION-II : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

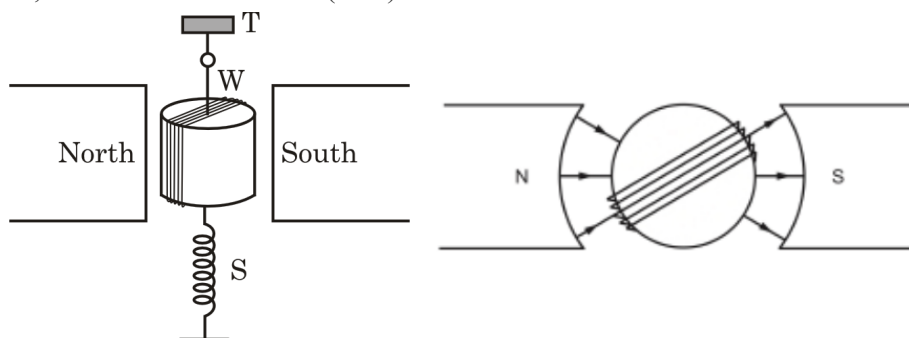
Zero Marks : 0 In all other cases.

1. The top and bottom faces of a rigid insulating box of dimensions $1\text{m} \times 1\text{m} \times 1\text{mm}$ and mass 1 kg are charge with uniform surface charge densities $+10^{-2}\text{ C/m}^2$ and -10^{-2} C/m^2 respectively and the box is placed in a uniform horizontal magnetic field $\vec{B} = (5\text{T})\hat{j}$. Assuming free space condition. Find velocity of box (in $\mu\text{m/s}$) when magnetic field is switched off.



2. Wind is blowing due east with speed 10 m/s . A car is moving west with speed 20 m/s towards a truck moving towards car with speed 10 m/s . If car blow horn of frequency 20 Hz , then find number of beats heard per second by a person in car. Assume waves reflected from truck reaches car. (speed of sound in air = 320 m/s)
3. A person can throw a ball to a maximum height of 20 m on the earth. If same person throw same ball on an imaginary planet whose density is twice of the earth then maximum possible radius of planet so that ball can escape from planet is $R\text{ km}$. Find R . (Radius of the Earth = $6.4 \times 10^6\text{ m}$)

4. A charged particle of charge $1\mu\text{C}$ and mass 10^{-6} kg is moving with a velocity of $9 \times 10^7 (\hat{i} + 2\hat{k})\text{ m/s}$ in space. There is a plane electromagnetic wave given by $\vec{E} = 10^2\text{ N/C} \hat{i} \sin(2\pi z - \omega t)$. The charge crosses origin at $t = \frac{5}{6}\text{ ns}$. At that moment, the magnitude of its acceleration is _____ m/s^2 (Neglect relativistic effects)
5. A galvanometer is not connected to any circuit. When the needle is slightly deflected, it undergoes undamped oscillations with angular frequency of 5 rad/s . When the galvanometer terminals are connected by a conducting wire, it undergoes damped oscillations with angular frequency 4 rad/s . The radial magnetic field is 0.3 T , area of coils is 100 cm^2 and there are 200 coils. If moment of inertia of assembly is $2.5 \times 10^{-3}\text{ kgm}^2$, estimate the resistance (in Ω) of the circuit.



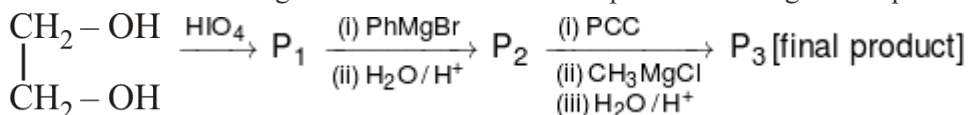
PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

<i>Full Marks</i>	: +4	If only (all) the correct option(s) is (are) chosen.
<i>Partial Marks</i>	: +3	If all the four options are correct but ONLY three options are chosen.
<i>Partial Marks</i>	: +2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
<i>Partial Marks</i>	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
<i>Zero Marks</i>	: 0	If none of the options is chosen (i.e. the question is unanswered).
<i>Negative Marks</i>	: -1	In all other cases.

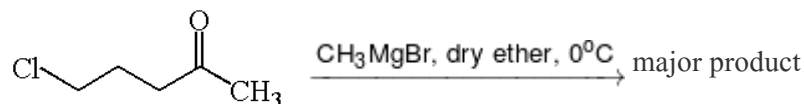
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Which of the following is/are correct for the final product of the given sequence of reaction.

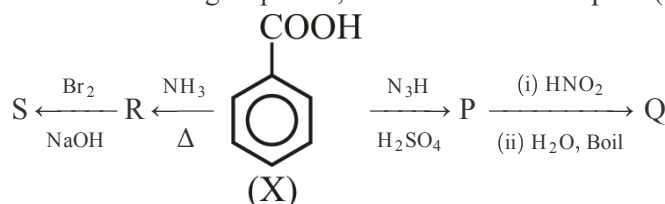


- (A) Compound P₃ on oxidation with (CrO₃ + H₂SO₄) gives a compound which can give positive 2,4-DNP test.
- (B) Compound P₃ on reaction with I₂ + NaOH gives yellow precipitate.
- (C) Compound P₃ can react with PCC.
- (D) Compound P₃ on reaction with MnO₂ gives carboxylic acid.

2. Which of the following is/are correct statement regarding major product?



- (A) Intramolecular nucleophilic substitution reaction takes place during the reaction sequence
 (B) 5 membered ring is formed as major product
 (C) Grignard reagents (CH_3MgBr) prefer to give nucleophilic addition on polar π -bond
 (D) Cyclic ether is formed as final major product
3. For the following sequence, choose the correct option(s) ?



- (A) (Q) will give positive Br_2 , H_2O test
 (B) (Q) on reaction with neutral FeCl_3 gives violet colour
 (C) Major product (P) & (S) are identical
 (D) Reaction sequence involves schmidt & Hoffmann bromamide degradation reaction
4. Select correct statements -
- (A) All adiabatic processes are isentropic
 (B) At equilibrium, at constant T & P, 'G' becomes minimum
 (C) At equilibrium, entropy of universe becomes maximum
 (D) G for a substance may change by changing T & P

5. Four solutions of various electrolytes are given alongwith their percent ionization
 (I) 0.1 M NaCl(aq.) (90 % ionized)
 (II) 0.05 M CaCl₂(aq.) (80 % ionized)
 (III) 0.04 M K₃[Fe(CN)₆](aq.) (60 % ionized)
 (IV) 0.03 M FeCl₃ (aq.) (70 % ionized)
 Identify correct statement(s) assuming all solutions to be dilute -
 (A) Solution (IV) has maximum vapour pressure
 (B) Solution (I) has minimum vapour pressure
 (C) Solution (III) has lowest freezing point
 (D) Solution (I) has highest boiling point
6. In the following reactions. The correct statements is/are :
 I. $\text{FeSO}_{4(\text{aq.})} + \text{NO} + \text{H}_2\text{O} \rightarrow \text{X}$
 II. Sodium nitroprusside_(aq.) + Na₂S → Y
 (A) Products X and Y both are paramagnetic
 (B) In the reaction I change of oxidation state of central atom occurs while in reaction II there is no change in oxidation state of metal
 (C) Product (X) can also be obtained when NO₂⁻ or NO₃⁻ react with dil. H₂SO₄ in presence of FeSO₄
 (D) Hybridisation of central atom of X is sp³d² and Y is d²sp³
7. 1L of 0.2 M HCl (aq.) is mixed with 1 L of 0.2 M weak monoprotic acid HA (K_a = 5 × 10⁻⁶) at 25°C. Select the correct options :
 (A) pH of the final solution will be 1.
 (B) % dissociation of HA in final solution is approximately 5 × 10⁻³ %
 (C) Conc. of A⁻ in final solution will be 1 × 10⁻³ M
 (D) Conc. of A⁻ in final solution will be 5 × 10⁻⁶ M

SECTION-I (ii) : (Maximum Marks: 18)

- This section contains **SIX (6)** questions of matching type.
- This section contains **TWO** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
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Paragraph for Questions 8 - 10

Column - I		Column - II		Column - III	
(I)		(i)	Cold dilute alkaline KMnO_4	(P)	Meso compound
(II)		(ii)	Oxymercuration demercuration	(Q)	Threo compound is formed
(III)		(iii)	Hydroboration Oxidation	(R)	Racemic Mixture
(IV)		(iv)	Br_2/CCl_4	(S)	Diastereo Isomers

8. Which of the following combination is **True** ?

- (A) III - iv - P (B) II - ii - Q
(C) I - i - R (D) IV - i - R

9. Which of the following combination is **Incorrect** ?

- (A) III - i - Q (B) III - i - R
(C) IV - i - P (D) II - ii - R

10. Which of the following is a case of syn addition?

- (A) III - i - R (B) III - ii - P
(C) IV - iv - P (D) IV - iv - R

Paragraph for Questions 11 - 13

Column-I Formula of oxide of nitrogen	Column-II Common method of preparation	Column-III Chemical nature & colour of oxide of nitrogen
(1) N_2O_3	(i) $HNO_3 + P_4O_{10} \rightarrow$	(P) neutral , colourless
(2) NO_2	(ii) $NO + N_2O_4 \xrightarrow{-30^\circ C}$	(Q) acidic , blue
(3) N_2O_5	(iii) $Pb(NO_3)_2 \xrightarrow{673K}$	(R) acidic , brown
(4) N_2O	(iv) $NH_4NO_3 \xrightarrow{\Delta}$	(S) acidic , colourless

11. Which of the following is correctly matched?
- (A) (1), (iv), (S) (B) (3), (iv), (P)
- (C) (4), (i), (R) (D) (2), (iii), (R)
12. Which of the following match is the CORRECT for N_2O_5 ?
- (A) (i), (R) (B) (iv), (S)
- (C) (i), (S) (D) None of these
13. Which of the following is NOT correctly matched :-
- (A) (1), (ii), (Q) (B) (4), (iii), (P)
- (C) (2), (iii), (R) (D) (4), (iv), (P)

SECTION-II : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
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Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

- For the reaction $A(g) \rightleftharpoons B(g)$; $K_p = 10$ at 350K. Calculate ΔG (in kcal/mole) at 350 K, if partial pressures of A & B gases are 10 bar & 1 bar respectively. $\left[\text{Given : } R = 2 \frac{\text{cal}}{\text{mol} - \text{K}} \right]$
- Using the data provided, calculate the multiple bond energy (kJ mol^{-1}) of a $C \equiv C$ in C_2H_2 . That energy is (take the bond energy of C-H bond as $350 \text{ kJ bond mol}^{-1}$).
 $2C(s) + H_2(g) \rightarrow C_2H_2(g)$; $\Delta H = 220 \text{ kJ/mol}$
 $2C(s) \rightarrow 2C(g)$; $\Delta H = 1400 \text{ kJ/mol}$
 $H_2(g) \rightarrow 2H(g)$; $\Delta H = 300 \text{ kJ/mol}$
 If the value of bond enthalpy is $x \text{ kJ/mol}$; calculate $\left(\frac{x}{1000} \right)$
- Salt 'X' $\xrightarrow{\text{dil. } H_2SO_4}$ No observation

$\downarrow$

$MnO_2 + \text{conc. } H_2SO_4 \xrightarrow{\Delta}$ Coloured gas/vapours

Total number of anions which will give coloured gas/vapour will be :-

$NO_2^\ominus$, $S_2O_3^{2-}$, Cl^- , Br^- , I^- , NO_3^- , F^-
- Aqueous $CuSO_4$ decolorizes on addition of KCN (excess) due to formation of complex (A).
 In complex "A". If
 (i) Number of d orbitals participating in hybridisation is / are 'a'
 (ii) Coordination number of Cu is "c".
 Then find $8a + 5c$.
- Mg crystallizes in HCP. Calculate total number of tetrahedral and octahedral voids in 12 milligrams of Mg ($N_A = 6 \times 10^{23}$) If your answer is $x \times 10^{20}$ then fill value of x in OMR sheet.

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 28)

- This section contains **SEVEN (07)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

<i>Full Marks</i>	:	+4	If only (all) the correct option(s) is (are) chosen.
<i>Partial Marks</i>	:	+3	If all the four options are correct but ONLY three options are chosen.
<i>Partial Marks</i>	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct.
<i>Partial Marks</i>	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
<i>Zero Marks</i>	:	0	If none of the options is chosen (i.e. the question is unanswered).
<i>Negative Marks</i>	:	-1	In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Let f, g and h are function differentiable on some open interval around 0 and satisfy the equation $f' = 2f^2gh + \frac{1}{gh}$, $f(0) = 1$, $g' = fg^2h + \frac{4}{fh}$, $g(0) = 1$ and $h' = 3fgh^2 + \frac{1}{fg}$, $h(0) = 1$. The function f is given by

- (A) $f(x) = 2^{-1/12} \left(\frac{\sin(6x + \frac{\pi}{4})}{\cos^2(6x + \frac{\pi}{4})} \right)^{1/6}$
- (B) $f(x) = 2^{-1/6} \left(\frac{\sin(6x + \frac{\pi}{4})}{\cos^2(6x + \frac{\pi}{4})} \right)^{1/12}$
- (C) $f(x) = (\sec 12x)^{1/12} (\sec 12x + \tan 12x)^{1/4}$
- (D) $f(x) = (\sec 12x)^{1/4} (\sec 12x + \tan 12x)^{1/12}$

2. A differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies following relations
- (i) $f(x) - f(y) = (x - y) f' \left(\frac{x + y}{2} \right), \forall x, y \in \mathbb{R}$
- (ii) $f(0) = f(1) = 0, f(2) > 0$
- Which of the following is/are true
- (A) $f(x)$ is an onto function
- (B) $f(x)$ has an extrema at $x = \frac{1}{2}$
- (C) $f(x)$ is a many one function
- (D) $\lim_{x \rightarrow -\infty} f(x) = -\infty$
3. Let A be the set of 2×2 matrices given by $A = [a_{ij}]$; where $a_{ij} \in \{0, 1, 2, 3, 4, 5\}$ such that $a_{11} + a_{12} + a_{21} + a_{22} = 5$, then
- (A) the number of matrices in A which are invertible are 32
- (B) the number of matrices in A which are symmetric or skew symmetric are 8
- (C) the number of matrices in A which are symmetric or skew symmetric such that $\text{Det } A$ is odd are 6
- (D) The total number of matrices in A are 56
4. $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}| = 1, |\vec{b}| = 4$ and $\vec{a} \cdot \vec{b} = 2$. If $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$, then which of the following is/are correct ?
- (A) $\vec{b} \cdot \vec{c} = 48$
- (B) $\vec{b} \cdot \vec{c} = -48$
- (C) Angle between $\vec{b}$ and $\vec{c}$ is $\frac{5\pi}{6}$
- (D) Angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{6}$
5. Consider a pyramid built on a square base of side 2 such that the lengths of the line segments joining its apex to the 4 corners of the base are equal and the height of the structure is equal to the length of a side of its base. Then which of the following statements is/are true
- (A) volume of the pyramid is $\frac{8}{3}$
- (B) the angle between the base and any other face is $\sin^{-1} \frac{2}{\sqrt{5}}$
- (C) the angle between two faces with common edge which passes through the apex is $\cos^{-1} \frac{1}{5}$
- (D) volume of the largest tetrahedron that can be inscribed in the pyramid is 1

6. If volume of a parallelepiped determined by the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ is 2, then volume of parallelepiped determined by the vectors

(A) $3(\vec{a} + \vec{b})$, $(\vec{b} + \vec{c})$ and $(\vec{c} + \vec{a})$ is 24

(B) $2(\vec{a} \times \vec{b})$, $3(\vec{b} \times \vec{c})$ and $(\vec{c} \times \vec{a})$ is 24

(C) $(\vec{a} - 2\vec{b})$, $(\vec{b} - 2\vec{c})$ and $(\vec{c} - 2\vec{a})$ is 14

(D) $(3\vec{a} - \vec{b})$, $(3\vec{b} - \vec{c})$ and $(3\vec{c} - \vec{a})$ is 52

7. Let $(1 + \sqrt{5})^n = \alpha_n + \beta_n \sqrt{5}$ where α_n and β_n are integers, then choose the correct option(s)

(A) $\lim_{\theta \rightarrow 0} \frac{\tan^{-1} \left(\frac{\alpha_{n+1} - \beta_{n+1}}{3} \right) \theta}{\sin^{-1} (\beta_n \theta)} = \frac{4}{3}$

(B) $\lim_{\theta \rightarrow 0} \frac{\tan^{-1} \left(\frac{\alpha_{n-1} - \beta_{n-1}}{3} \right) \theta}{\sin^{-1} (\beta_n \theta)} = \frac{3}{4}$

(C) $\left(\frac{\alpha_n}{2^n} \right)^2 - 5 \left(\frac{\beta_n}{2^n} \right)^2 = (-1)^n$

(D) $\left(\frac{\alpha_n}{2^n} \right)^2 - 5 \left(\frac{\beta_n}{2^n} \right)^2 = (-1)^{\frac{n}{2}}$

SECTION-I (ii) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions of matching type.
- This section contains **TWO** tables (each having 3 columns and 4 rows)
- Based on each table, there are **THREE** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : -1 In all other cases

Paragraph for Questions 8 - 10

Column-I : $y = f(x)$ with respective domain.

Column-II : Interval in which $f(x)$ is increasing

Column-III : Interval in which $f(x)$ is decreasing

Column-I		Column-II		Column-III	
(I)	$y = \sin x + \cos x$ in $\left(0, \frac{\pi}{2}\right)$	(i)	$\left(0, \frac{\pi}{4}\right)$	(P)	$\left(0, \frac{\pi}{4}\right)$
(II)	$y = \sin 2x$ in $\left(0, \frac{\pi}{2}\right)$	(ii)	$\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$	(Q)	$\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$
(III)	$y = \sin^5 x + \cos^5 x - 1$ in $\left(0, \frac{\pi}{2}\right)$	(iii)	$\left(0, \frac{\pi}{3}\right)$	(R)	$\left(0, \frac{1}{e}\right)$
(IV)	$y = x^x$ in $\left(0, \frac{\pi}{2}\right)$	(iv)	$\left(\frac{1}{e}, \frac{\pi}{2}\right)$	(S)	$\left(\frac{1}{e}, \frac{\pi}{2}\right)$

8. Which of the following options is the only **CORRECT** combination ?

- (A) (I) (i) (S) (B) (II) (iii) (S)
(C) (III) (ii) (P) (D) (I) (ii) (P)

9. Which of the following options is the only **CORRECT** combination ?

- (A) (I) (iii) (P) (B) (II) (ii) (P)
(C) (III) (iii) (P) (D) (IV) (iv) (R)

10. Which of the following options is the only **CORRECT** combination ?

- (A) (I) (iv) (R) (B) (II) (i) (Q)
(C) (III) (iii) (Q) (D) (II) (ii) (P)

Paragraph for Questions 11 - 13

Answer Q.11, Q.12 and Q.13 by appropriately matching the information given in the three columns of the following table

Column 1		Column 2		Column 3	
(I)	$\int_0^{\pi/2} \sin^8 x \, dx = \frac{A\pi}{2^B}$ (where A and 2^B are co-prime numbers), then $\frac{A}{B-1}$ is	(i)	If slope of tangent to the curve $y = x + \sin x \cos x$ at $x = \frac{\pi}{2}$ is m, then $(m+7)$ is	(P)	odd
(II)	$\frac{\int_0^8 [x] \, dx}{\int_0^8 \{x\} \, dx}$ is (where $[.]$ is greatest integer function and $\{.\}$ fractional part function respectively)	(ii)	If y-intercept of perpendicular bisector of $P(2,5)$ and $Q(k,2)$ is $\frac{3}{2}$, then $k(k > 0)$ can be	(Q)	even
(III)	If $\ell = \lim_{t \rightarrow 0} \frac{1}{t^4} \int_0^t \frac{x^2 \ln(1+x)}{x^7+3} \, dx$ then $\frac{3}{4\ell}$ is	(iii)	Area of polygon formed by $x^2 + x - 2 = 0$ and $y^2 - 5y + 4 = 0$ is	(R)	prime
(IV)	$\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{2n}} + \frac{1}{\sqrt{3n}} + \dots + \frac{1}{2n} \right)$ is	(iv)	Rate of change of area of an equilateral triangle w.r.t. its side when side is $\frac{10}{\sqrt{3}}$ is	(S)	composite

11. Which of the following options is the only **CORRECT** combination ?

- (A) (I) (iii) (Q,R) (B) (I) (i) (P,R)
(C) (I) (iv) (P,R) (D) (I) (ii) (Q,R)

12. Which of the following options is the only **CORRECT** combination ?

- (A) (II) (i) (P,R) (B) (II) (ii) (P,R)
(C) (II) (iii) (Q,S) (D) (II) (iv) (Q,S)

13. Which of the following options is the only **CORRECT** combination ?

- (A) (III) (ii) (P,S) (B) (III) (i) (Q,S)
(C) (IV) (ii) (Q,S) (D) (IV) (iii) (P,S)

SECTION-II : (Maximum Marks: 15)

- This section contains **FIVE (05)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. If in a 4×4 matrix the sum of elements of each row, column and both the main diagonals is 1, then sum of the four corner elements is equal to
2. Let $A = \begin{bmatrix} 5 & 17 \\ 13 & 29 \end{bmatrix}$ and B is 2×2 matrix whose elements are elements of set $P\{3,5,7,11,13,17,19,23,29\}$.
If all the elements of AB are multiple of 4, then the total number of possible matrices B with distinct elements.
3. Consider a 3×3 matrix $A = \{a_{ij}\}$ where $a_{ij} = \frac{1}{i+j-1} \forall i, j \in \{1, 2, 3\}$. If matrix $B = \{b_{ij}\}$ of order 3×3 exists such that $A^{-1} = 3B$, then the value of $\frac{|B|}{a_{23} \cdot b_{22}}$ is (where $|X|$ denotes determinant value of matrix X)
4. In a ΔABC , $AB = 4$, $AC = 5$, O is circumcenter of ΔABC , then the value of $2\overline{AO} \cdot \overline{BC}$?
5. If $a_n = \left(\frac{2n+1}{n+1}\right)^8 C_n$; $n = \{0, 1, 2, \dots, 7, 8\}$
 $b_n = \left(\frac{3n+1}{n+1}\right)^7 C_n$; $n = \{0, 1, 2, \dots, 7\}$
If $a_p = \max_{0 \leq n \leq 8} (a_n)$ and $b_q = \min_{0 \leq n \leq 7} (b_n)$, where $n \in W$
then the value of $(p + q)$ is

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

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